

# On Edge Collapse Of Random Simplicial Complexes

Soumik Dutta, Joint work with Jean-Daniel Boissonnat, Kunal Dutta, and Siddharth Pritam

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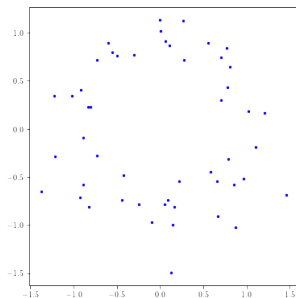
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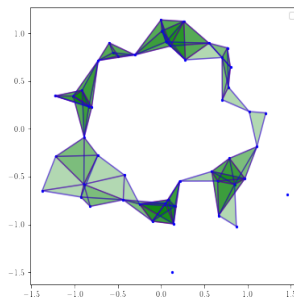
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# Introduction

- ▶ Topological Data Analysis (TDA) is a newly emerging field of data analysis in recent decades. TDA aims to extract topological features, such as **persistent diagrams** and **Betti numbers**, from spatially distributed data.
- ▶ First, the dataset is turned into a topological object suitable for computation.
- ▶ Then **homological information** and **persistent homology (PH)** are extracted.



(a) Point cloud



(b) Vietoris-Rips complex

# Simplicial Complex

- ▶ Simplicial Complex is a discrete version of topological spaces.
- ▶ They are made of vertices and **faces/simplices** (an edge being a face of dimension 1) of different dimensions satisfying that the intersection of two faces is either another face or empty.
- ▶ An abstract simplicial complex is a collection of faces that is closed under subset operation.
- ▶ They may contain **holes, voids, and higher-dimensional analogs of them.**

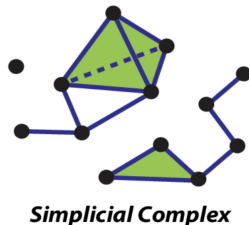
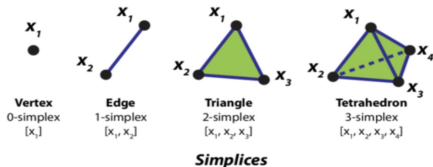


Image source: Topological portraits of multiscale coordination dynamics. DOI: 10.1016/j.jneumeth.2020.108672

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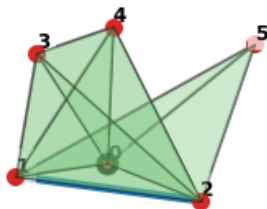
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- ▶ **Edge collapse** was introduced by **J-D. Boissonnat and S. Pritam, SoCG'20**. In this process, **edges are removed**.

# Edge Collapse

- ▶ An edge  $e$  in the simplicial complex  $K$  is dominated by another vertex  $v \in K - e$  if all the maximal simplices of  $K$  that contain  $e$  also contain  $v$ .
- ▶ Removing the dominated edge along with all its cofaces does *not* change the homotopy type of the complex. The process is called edge collapse.
- ▶ Two topological spaces have the same homotopy type if one space can be continuously retracted onto the other.

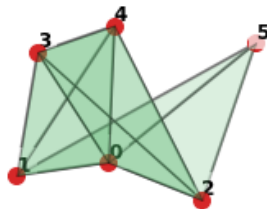
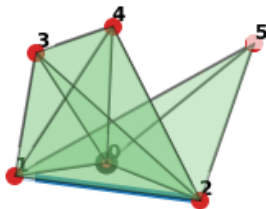
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# Motivations

- ▶ The number of  $d$ -simplices is typically  $O(m^{d-2})$  ( $m = \#$  edges) and time complexity of persistent homology algorithm is  $O(N^\omega)$  ( $N = \#$  simplices,  $\omega =$  matrix multiplication exponent).

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- ▶ Observed experimental and practical successes have led to its deployment in software packages like GUDHI.
- ▶ Our study is a first step in the theoretical understanding in that direction.
- ▶ Our results also provide an average case analysis for edge collapse when no extra information about the data set is provided.

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- ▶ The [random clique complex model \( \$X\(n, p\)\$ \)](#) was introduced by [Matthew Kahle in 2009](#). In the Erdos-Renyi model of random graph,  $G(n, p)$ , every  $k$ -clique is replaced with a  $k - 1$  face.

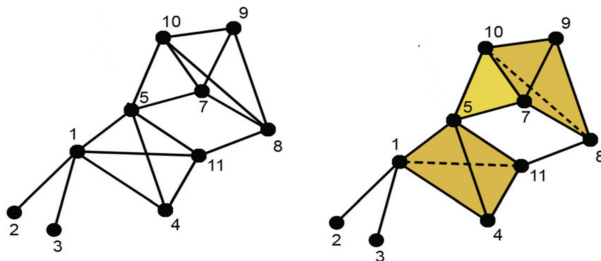


Image source: Persistent Homology of Complex Networks. DOI: 10.1088/1742-5468/2009/03/P03034

Our goal is to study the effect of edge collapse in reducing the size of the complex for random clique complexes.

# t-core Algorithm

We shall iteratively run the edge collapse procedure according to the following algorithm.

---

**Algorithm 1:** Compute  $t$ -core under edge collapse

---

**Data:** Simplicial complex  $X$

**Result:**  $R_t(X)$ , i.e.,  $t$ -core of  $X$  under edge collapse

- 1 Calculate the set  $D$  of all the dominated edges of the current complex;
  - 2 Shuffle elements of  $D$  uniformly randomly;
  - 3 **for**  $e$  in  $D$  **do**
  - 4     | If  $e$  is dominated by some vertex in the current complex remove  $e$  and all its cofaces.
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**Algorithm 2:** Compute  $t$ -core under edge collapse

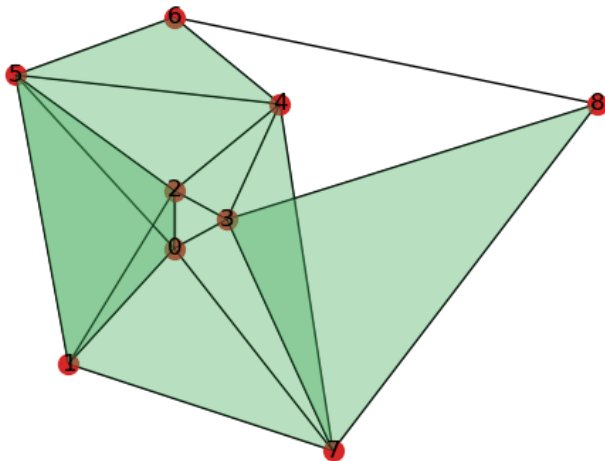
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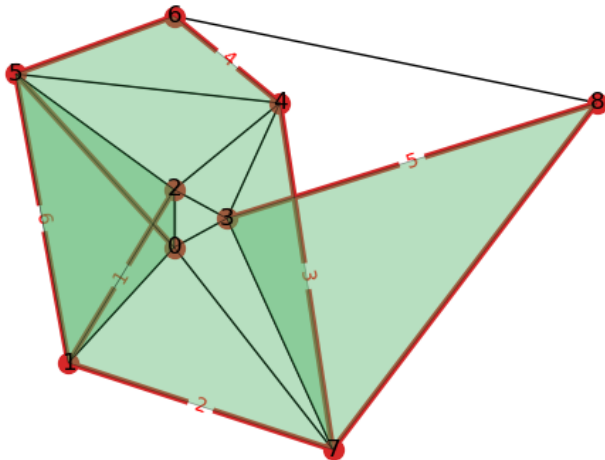
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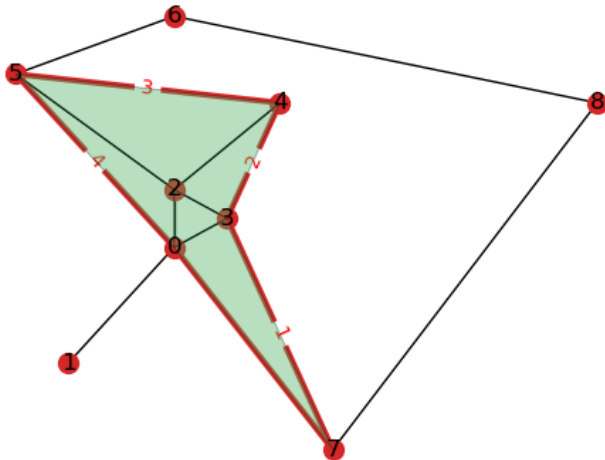
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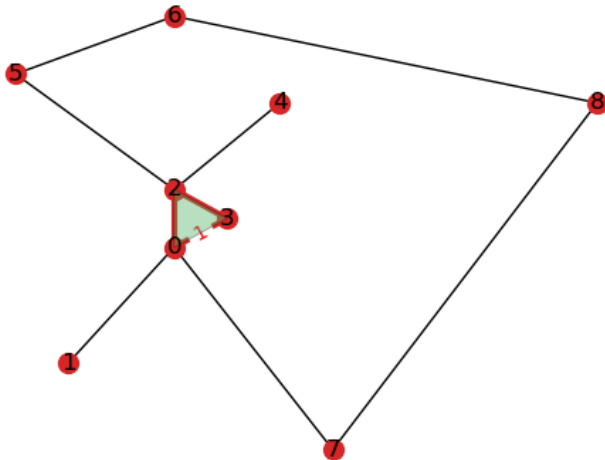
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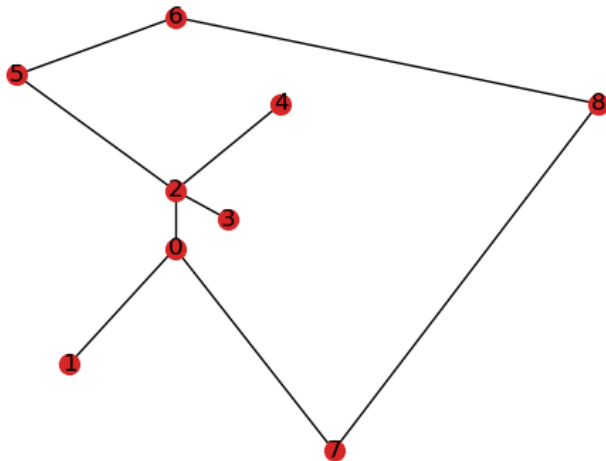
The  $\infty$ -core is simply called (final) core.











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- ▶ Define  $\eta_2 = \inf\{\eta \mid x = e^{-\eta(1-x)^2} \text{ has a solution in } (0, 1)\}$ . In particular,  $\eta_2 \approx 2.45$  and  $c = \sqrt{\eta_2} \approx 1.57$

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- ▶ By  $t - \text{core}(\cdot)$  we mean the remaining complex after  $t$  phases of edge collapse. By  $\text{core}(\cdot)$  we mean the remaining complex after the maximal number of edge collapsing phases.

# Results

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## Theorem

Define  $\eta_2 = \inf\{\eta | x = e^{-\eta(1-x)^2} \text{ has a solution in } (0, 1)\}$ . Let  $X \sim X(n, p)$  with  $p = c/\sqrt{n}$  and  $c > \sqrt{\eta_2}$ . Let  $\gamma$  be the smallest positive solution of the equation  $x = e^{-c^2(1-x)^2}$ . Then a.a.s.

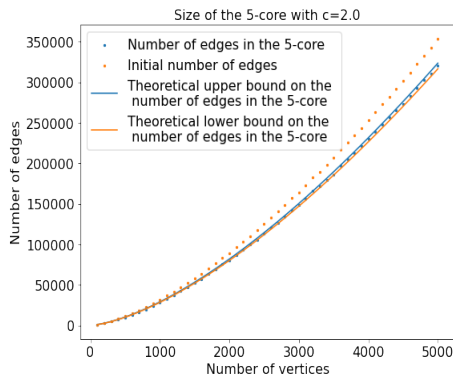
$$e(\text{core}(X)) \leq (1 + o(1))\mathbb{E}[e(X)](1 - (c^2/3)(1 - (1 - \gamma)^3)) \quad (1)$$

and

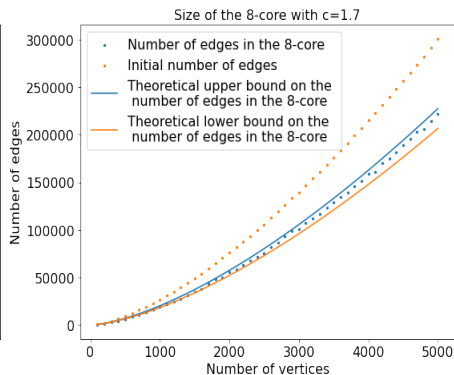
$$(1 + o(1))\mathbb{E}[e(X)](1 - \gamma - c^2\gamma(1 - \gamma)^2) \leq e(\text{core}(X))$$

# Simulation

Code: <https://github.com/soumikdt/Edge-Collapse-On-Random-Clique-Complex>

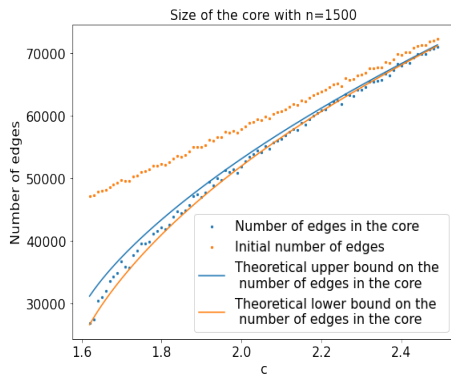


(a) Size of the 5-core with  $c = 2.0$ .

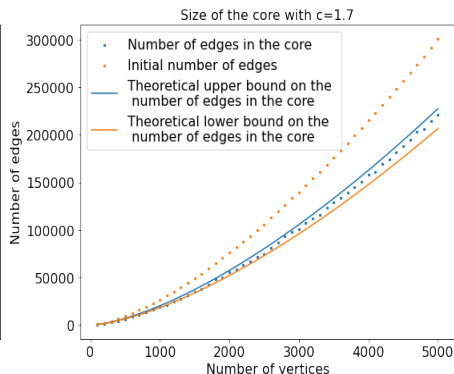


(b) Size of the 8-core with  $c = 1.7$ .

**Figure:** Fig. 3a shows the number of edges in the 5-core of  $X \sim X(n, 2/\sqrt{n})$  along with the initial number of edges. The solid line indicates the theoretical upper and lower bounds on the expected number of edges. Fig. 3b shows the same thing for the 8-core of  $X \sim X(n, 1.7/\sqrt{n})$ .



(a) Size of the core plotted against  $c$ .



(b) Upper and lower bound of the size of the core.

**Figure:** In Fig. 4a, we plotted the number of edges in the core of  $X \sim X(1500, c/\sqrt{1500})$  for  $c \in [1.6, 2.5]$  along with the initial number of edges. The solid lines indicate the theoretical upper and lower bounds on the expected number of edges. Note that these bounds tend to coincide as  $c$  increases as shown in the figure. In Fig. 4b, we ran the pruning procedure until there were no more dominated edges. The figure shows the number of edges in the final core of  $X \sim X(n, 1.7/\sqrt{n})$  along with the initial number of edges. The solid line indicates the theoretical upper bound on the expected number of edges.

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## Outline of the proof:

- ▶ Upper and lower bound on  $t$ -core.
  - Expectation
    - We first identify the local structure.
    - We then exploit the tree like property of the local structure.
  - Concentration
    - We first identify a class of critical structures responsible for large deviation from the expectation.
    - Then we apply a concentration inequality that we developed for typically bounded functions.
- ▶ Upper bound on core.
- ▶ lower bound on core.

# The Local Structure: Random 2–tree

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- ▶ Thus the complex locally looks like a 2-dimensional tree.

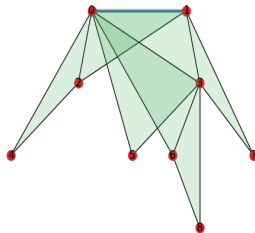


Figure: 2-dimensional Tree

- ▶ This holds with high probability for  $O(\log n / \log \log n)$  depth around any edge.

- ▶ Each edge has  $\text{Binomial}(c^2/n, n-2) \approx \text{Poisson}(c^2)$  children.

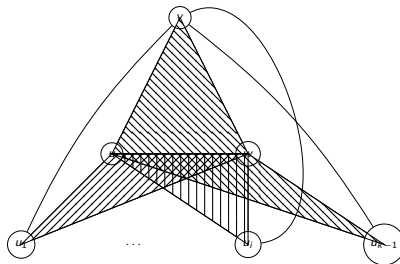
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- ▶ This gives us bounds on the expected size of the core.

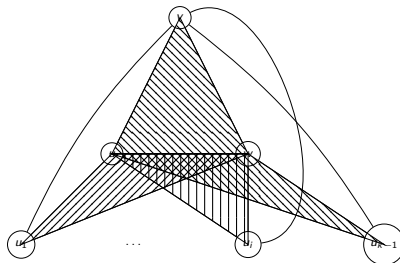
# Concentration of $t$ -core size: Identifying the critical structures

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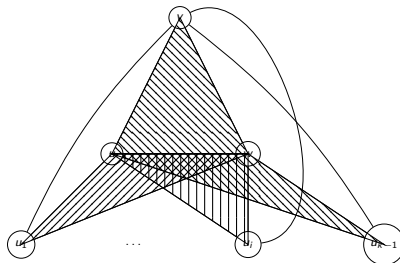
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- ▶ Thus the actual size of the  $t$ -core can drastically vary from its expected size depending on whether critical complexes are present or not.
- ▶ We prove that a critical complex must be present as a subcomplex for large deviation from the expected size to occur.

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### Theorem

Let  $X \sim X(n, p)$ . Then for any  $s \geq 0$ , we have,

$$\Pr\{|e(t - \text{core}(X)) - \mathbb{E}[e(t - \text{core}(X))]| \geq s \cdot n\} \leq 2 \exp(-s^2 \Theta(\sqrt{n})) + O(1/n^{5.5})$$

# Upper bound of core

- ▶ First we observe that, for a fixed flag complex  $X$ ,  $e(t - \text{core}(X)) \geq e((t + 1) - \text{core}(X))$  for any  $t \geq 1$ .

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- ▶ Thus  $e(t - \text{core}(X)) \geq e(\text{core}(X))$  for any  $t \geq 1$ .
- ▶ In particular, we shall take  $t = \log n / 16 \log \log n$ .

# Upper bound of core

- ▶ First we observe that, for a fixed flag complex  $X$ ,  $e(t - \text{core}(X)) \geq e((t+1) - \text{core}(X))$  for any  $t \geq 1$ .
- ▶ Thus  $e(t - \text{core}(X)) \geq e(\text{core}(X))$  for any  $t \geq 1$ .
- ▶ In particular, we shall take  $t = \log n / 16 \log \log n$ .
- ▶ This gives the desired upper bound.

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- ▶ Due to this recursive nature of the property  $\mathcal{A}$ , an edge with property  $\mathcal{A}$  never gets collapsed a.a.s.
- ▶ Thus by studying  $\mathcal{A}$ , which is a property defined on a local neighborhood, we can infer about a global property.

THANK YOU

A full version of the paper is available at  
[https://www.researchgate.net/publication/379119064\\_On\\_Edge\\_Collapse\\_of\\_Random\\_Simplicial\\_Complexes](https://www.researchgate.net/publication/379119064_On_Edge_Collapse_of_Random_Simplicial_Complexes)